SUPPLEMENTARY MATERIAL

VIII. NOTATIONS AND MATHEMATICAL DEFINITIONS

In this section, we introduce some symbol definitions and the preliminary knowledge, and supplement the specific definitions of related concepts mentioned in Section II.

A. Definitions of norm

We first introduce the definitions of norm. Given a matrix $A \in \mathbb{R}^{m_1 \times m_2}$, let $a_{i_1 i_2}$ denote the (i_1, i_2)-th component of A, then the Frobenius norm of A is defined as

$$\|A\|_F := \sqrt{\sum_{i_1=1}^{m_1} \sum_{i_2=1}^{m_2} a_{i_1 i_2}^2}.$$

then the ℓ_1-norm of A is defined as

$$\|A\|_1 := \sum_{i_1=1}^{m_1} \sum_{i_2=1}^{m_2} |a_{i_1 i_2}|.$$

B. Mathematical definitions for nonconvex analysis

We first define $\langle \cdot, \cdot \rangle$ as the inner product.

Next, we recall several basic concepts for nonconvex analysis [22], [17].

Definition 1. *Let $\boldsymbol{S}$ be a set, the derived set $\boldsymbol{S}'$ is defined as the set of overall cluster points of $\boldsymbol{S}$.*

Proposition 1. *A bounded closed set in finite-dimensional space is a compact set.*

Definition 2. *A function $g : \mathbb{R}^n \to (-\infty, +\infty]$ is said to be proper if $\mathrm{dom}\ g \neq \emptyset$, where $\mathrm{dom}\ g = \{x \in \mathbb{R} :\ g(x) < \infty\}$.*

Definition 3. *If $\forall x_k \in \mathrm{dom}\ f$, we have*

$$\lim_{k\to\infty} x_k = x,\ f(x) \leq \liminf_{k\to\infty} f(x_k)\ ,$$

then f is said to be lower semicontinuous on $\mathrm{dom}\ f$.

Definition 4. *Let f be a proper lower semicontinuous function, the Fréchet subdifferential of f at x, written $\hat{\partial} f(x)$, is the set of all vectors u which satisfy*

$$\lim_{y \neq x, y \to x} \cdot \frac{f(y) - f(x) - \langle u,\ y - x \rangle}{\|y - x\|} \geq 0,$$

when $x \notin \mathrm{dom}\ f$, then set $\hat{\partial} f(x) = \emptyset$.

Proposition 2. *If f is a proper lower semicontinuous function and has a local minimum at x^*, then $0 \in \partial f(x^*)$.*

Proposition 3. *Let f be a proper lower semicontinuous function, and g be a continuously differential function, then we have $\forall x \in \mathrm{dom}\ f$, $\partial(f + g)(x) = \partial f(x) + \nabla g(x)$.*

Proposition 4. *Set f: $R^n \to R$, if $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$, then $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2$, where L is the Lipschitz constant of ∇f.*

Definition 5. *Desingularization function: we denote the desingularization function $\phi : [0, \eta) \to \mathbb{R}_+$ which satisfies these conditions: (i) $\phi(0) = 0$. (ii) ϕ is continuously differentiable on $[0, \eta)$, and ϕ is a concave function from in $(0, \eta)$. (iii) $\phi(s)' > 0$ $(\forall s \in (0, \eta))$.*

Definition 6. *Let f be a proper lower semicontinuous function, f is said to have the KŁ property on $\bar{u} \in dom(\partial f)$, if there exists $\eta \in (0, +\infty]$, and U is the neighborhood of $\bar{u}$, any u in U that satisfies the condition $f(\bar{u}) < f(u) < f(\bar{u}) + \eta$ has the following inequalities established.*

$$\phi'(f(u) - f(\bar{u}))dist(0, \partial f(u)) > 1,$$

where ϕ satisfy (Definition 5). If f has the KŁ property at each point of $\mathrm{dom}\ \partial f$, then f is a KŁ function.

Proposition 5. *(Uniformized KŁ property) Let Ω be a compact set and f: $\mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ be a proper and lower semicontinuous function. Assume that f is constant on Ω and is a KŁ function. Then, there exist $\epsilon > 0$, $\eta > 0$ such that $\forall x \in \Omega$ and all x in the intersection $\{x \in \mathbb{R}^n : dist(x, \Omega) < \epsilon\} \cap \{f(\bar{x}) < f(x) < f(\bar{x}) + \eta\}$ one has*

$$\phi'(f(x) - f(\bar{x}))dist(0, \partial f(x)) > 1.$$

IX. CONVERGENCE ANALYSIS

In this section, we prove all the theorems in Section V. Notably, to simplify the proof in this section, we have already defined the sequence generated by Algorithm 1 in the main text as

$$z^k = A_{(N)}^k,\ c^k = Y_{(N)}^k. \tag{10}$$

A. Proof of Theorem 1

Proof. Because

$$\begin{aligned}
&\nabla_{A_i} H(\{A_j\}_{j=1}^{i-1}, B_i, \{A_j\}_{j=i+1}^{N}) \\
&- \nabla_{A_i} H(\{A_j\}_{j=1}^{i-1}, C_i, \{A_j\}_{j=i+1}^{N}) \\
&= ((\prod_{j=1}^{i-1} A_j)^T \prod_{j=1}^{i-1} A_j)(B_i - C_i)((\prod_{j=i+1}^{N} A_j)(\prod_{j=i+1}^{N} A_j)^T).
\end{aligned}$$

From the Frobenius norm is a self-consistent matrix norm, which conforms to the norm definition of a linear operator [22], thus we have

$$\begin{aligned}
&\|\nabla_{A_i} H(\{A_j\}_{j=1}^{i-1}, B_i, \{A_j\}_{j=i+1}^{N}) \\
&- \nabla_{A_i} H(\{A_j\}_{j=1}^{i-1}, C_i, \{A_j\}_{j=i+1}^{N})\|_F \\
&= \|((\prod_{j=1}^{i-1} A_j)^T \prod_{j=1}^{i-1} A_j)(B_i - C_i)(\prod_{j=i+1}^{N} A_j)(\prod_{j=i+1}^{N} A_j)^T\|_F \\
&\leq \|(\prod_{j=1}^{i-1} A_j)^T \prod_{j=1}^{i-1} A_j\|_F \\
&\times \|(B_i - C_i) \prod_{j=i+1}^{N} A_j(\prod_{j=i+1}^{N} A_j)^T)\|_F \\
&\leq \|(\prod_{j=1}^{i-1} A_j)^T \prod_{j=1}^{i-1} A_j\|_F \times \| \prod_{j=i+1}^{N} A_j(\prod_{j=i+1}^{N} A_j)^T)\|_F \\
&\times \|(B_i - C_i)\|_F.
\end{aligned}$$

The Theorem 1 is true. □

B. Proof of Theorem 2

Proof. (i) If $J(A_{(N)}^{k+1}) \leq J(A_{(N)}^k) - \rho_t \|A_{(N)}^{k+1} - Y_{(N)}^k\|_F^2$, then the Theorem 2(i) is true. If not, since Theorem 1, we know that $H(A_{(N)})$ satisfies Proposition 4. Thus, we have

$$\begin{aligned}
H(A_1^{k+1}, \{A_i^k\}_{i=2}^N) &\leq \langle \nabla_{A_1} H(A_{(N)}^k), A_1^{k+1} - A_1^k \rangle \\
&+ \frac{L_{\nabla_{A_1^k} H}}{2} \|A_1^{k+1} - A_1^k\|_F^2 + H(A_{(N)}^k).
\end{aligned} \tag{11}$$

From Eq. (9), we can also obtain

$$F_1(A_1^k) \geq F_1(A_1^{k+1}) + \frac{1}{2\sigma_1^k}\|A_1^{k+1} - A_1^k\|_F^2 + \langle \nabla_{A_1} H(A_{(N)}^k), A_1^{k+1} - A_1^k \rangle. \quad (12)$$

Then sum of the Eq. (11) and Eq. (12), we have

$$H(A_1^{k+1}, \{A_i^k\}_{i=2}^N) + F_1(A_1^{k+1}) \leq H(A_{(N)}^k) + F_1(A_1^k) - \rho\|A_1^{k+1} - A_1^k\|_F^2, \quad (13)$$

where $\rho = \min(\frac{1}{2\sigma_1^k} - \frac{L_{\nabla_{A_1^k} H}}{2})$. Assuming Theorem 2(i) holds when $n = N-1$, i.e.,

$$H(A_{(N-1)}^{k+1}, A_N^k) + \sum_{i=1}^{N-1} F_i(A_i^{k+1}) \leq H(A_{(N)}^k) + \sum_{i=1}^{N-1} F_i(A_i^k) - \rho\|A_{(N-1)}^{k+1} - A_{(N-1)}^k\|_F^2, \quad (14)$$

where $\rho = \min(\{\frac{1}{2\sigma_i^k} - \frac{L_{\nabla_{A_i^k} H}}{2}\}_{i=1}^{N-1})$. Similarly, from Eq. (9), we infer

$$F_N(A_N^k) \geq F_N(A_N^{k+1}) + \frac{1}{2\sigma_N^k}\|A_N^{k+1} - A_N^k\|_F^2 + \langle \nabla_{A_N} H(A_{(N-1)}^{k+1}, A_N^k), A_N^{k+1} - A_N^k \rangle. \quad (15)$$

From Theorem 1 and Proposition 4, we also have

$$\begin{aligned} H(A_{(N)}^{k+1}) &\leq H(A_{(N-1)}^{k+1}, A_N^k) \\ &+ \langle \nabla_{A_N} H(A_{(N-1)}^{k+1}, A_N^k), A_N^{k+1} - A_N^k \rangle \\ &+ \frac{L_{\nabla_{A_N^k} H}}{2}\|A_N^{k+1} - A_N^k\|_F^2. \end{aligned} \quad (16)$$

Then sum of the Eq. (14), Eq. (15) and Eq. (16), we have

$$J(A_{(N)}^{k+1}) \leq J(A_{(N)}^k) - \rho\|z^{k+1} - c^k\|_F^2, \quad (17)$$

where $\rho = \min(\{\frac{1}{2\sigma_i^k} - \frac{L_{\nabla_{A_i^k} H}}{2}\}_{i=1}^N)$. This shows the Theorem 2(i) is true.

(ii) We know that $\frac{\alpha}{2}\operatorname{Tr}((A_N)^T L A_N)$ in Eq. (7) is nonnegative. Hence, Eq. (7) is nonnegative. From Eq. (17), we have

$$\rho\|z^{k+1} - c^k\|_F^2 \leq J(A_{(N)}^k) - J(A_{(N)}^{k+1}).$$

Sum of both side, since Eq. (7) is nonnegative, thus we have

$$\begin{aligned} \rho\sum_{k=1}^{\infty}\|z^{k+1} - c^k\|_F^2 &\leq \sum_{k=1}^{\infty}(J(A_{(N)}^k) - J(A_{(N)}^{k+1})) \\ &= J(A_{(N)}^1) - \inf J \\ &< \infty. \end{aligned}$$

It follows that

$$\lim_{k\to\infty}\|z^{k+1} - c^k\|_F^2 = 0.$$

□

C. Lemma: subgradient lower bound property of Algorithm 1

Before continuing with the proof of the theorems of Section V, we prove the following lemma.

Lemma 1. *If we define*

$$\begin{aligned} f_i^{k+1} &= \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) + \frac{1}{\sigma_i^k}(Y_i^k - A_i^{k+1}) \\ &- \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N). \end{aligned} \quad (18)$$

Then we have $\|\{f_i^{k+1}\}_{i=1}^N\|_F \leq \rho_b\|z^{k+1} - c^k\|_F$, *where* ρ_b *is a positive constant.*

Proof. By Proposition 2 and Proposition 3, Eq. (9) follows that

$$0 \in \partial_{A_i} F_i(A_i^{k+1}) - \frac{1}{\sigma_i^k}(Y_i^k - A_i^{k+1}) + \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N). \quad (19)$$

Thus we also have

$$\begin{aligned} f_i^{k+1} &= \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) - \frac{1}{\sigma_i^k}(A_i^{k+1} - Y_i^k) \\ &- \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N) \\ &\in \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, A_i^{k+1}, \{A_j^k\}_{j=i+1}^N) \\ &+ \partial_{A_i}(\sum_{j=1}^i F_j(A_j^{k+1}) + \sum_{j=i+1}^N F_j(A_j^k)) \\ &= \partial_{A_i} J(\{A_j^{k+1}\}_{j=1}^{i-1}, A_i^{k+1}, \{A_j^k\}_{j=i+1}^N). \end{aligned} \quad (20)$$

Since Theorem 1, we have

$$\begin{aligned} \|f_i^{k+1}\|_F &= \|\nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) + \frac{1}{\sigma_i^k}(Y_i^k - A_i^{k+1}) \\ &- \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N)\|_F, \\ &\leq \|\nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) \\ &- \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N)\|_F \\ &+ \|\frac{1}{\sigma_i^k}(A_i^{k+1} - Y_i^k)\|_F \\ &\leq (L_{\nabla_{A_i^k} H} + \frac{1}{\sigma_i^k})\|A_i^{k+1} - Y_i^k\|_F. \end{aligned} \quad (21)$$

Thus we infer

$$\|\{f_i^{k+1}\}_{i=1}^N\|_F \leq \rho_b\|z^{k+1} - c^k\|_F, \quad (22)$$

where $\rho_b = \max(\{\frac{1}{\sigma_i^k} + L_{\nabla_{A_i^k} H}\}_{i=1}^N)$. □

D. Proof of Theorem 3

Proof. (i) Observe that z' can be viewed as an intersection of compact sets

$$z' = \cap_{q\in\mathbb{N}}\overline{\cup_{k\geq q}\{z^k\}}.$$

From proposition 1, z' is compact.

(ii) $\forall \overline{z} \in z'$, there exists a subsequence z^{k_j} such that

$$\lim_{j\to\infty} z^{k_j} = \overline{z}.$$

Let

$$F(z^{k_j}) = \sum_{i=1}^N F_i(A_i^{k_j}). \quad (23)$$

Since F_i is lower semicontinuous [17], from Definition 3 and Eq. (23), we obtain that

$$\lim_{j\to\infty}\inf F(z^{k_j}) \geq F(\overline{z}). \quad (24)$$

Choosing $k = k_j - 1$, from Eq. (9), we infer

$$\lim_{j\to\infty} \sup F_i(A_i^{k_j}) \le \lim_{j\to\infty} \sup(F_i(\overline{A_i}) + \frac{1}{2\sigma_i^k}\|\overline{A_i} - Y_i^k\|_F^2 + \langle \nabla_{A_i} H(\{A_n^{k+1}\}_{n=1}^{i-1}, Y_i^k, \{A_n^k\}_{n=i+1}^N), \overline{A_i} - Y_i^k\rangle), \quad (25)$$

where $\lim_{j\to\infty} A_i^{k_j} = \overline{A_i}$. Since $\lim_{j\to\infty} z^{k_j} = \overline{z}$ and $\lim_{j\to\infty} \|z^{k_j} - c^{k_j-1}\|_F = 0$ (Theorem 2(ii)), we can get that

$$\begin{aligned}\lim_{j\to\infty} \|\overline{A_i} - Y_i^k\|_F &\le \lim_{j\to\infty} \|\overline{A_i} - A_i^{k_j}\|_F \\ &+ \lim_{j\to\infty} \|A_i^{k_j} - Y_i^{k_j-1}\|_F \\ &= 0, \end{aligned} \quad (26)$$

From Eq. (25) and Eq. (26), we infer

$$\lim_{j\to\infty} \sup F_i(A_i^{k_j}) \le \lim_{j\to\infty} \sup F_i(\overline{A_i}), \quad (27)$$

where $\lim_{j\to\infty} A_i^{k_j} = \overline{A_i}$. Therefore, from Eq. (23) and Eq. (27), we have

$$\lim_{j\to\infty} \sup F(z^{k_j}) \le \lim_{j\to\infty} \sup F(\overline{z}). \quad (28)$$

From Eq. (23), Eq. (28) and Eq. (24), we infer

$$\lim_{j\to\infty} F(z^{k_j}) = F(\overline{z}). \quad (29)$$

Since Theorem 2, we have

$$\lim_{j\to\infty} H(z^{k_j}) = H(\overline{z}). \quad (30)$$

Thus from Eq. (29) and Eq. (30), we infer

$$\begin{aligned}\lim_{j\to\infty} H(z^{k_j}) + \lim_{j\to\infty} F(z^{k_j}) &= \lim_{j\to\infty} (H(z^{k_j}) + F(z^{k_j})) \\ &= \lim_{j\to\infty} J(z^{k_j}) \\ &= J(\overline{z}).\end{aligned}$$

This means J is constant on z'.

(iii) From Eq. (20), we know that

$$f_i^{k+1} \in \partial_{A_i} J(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N), \quad (31)$$

where the definitions of f_i^{k+1} is the same as in Eq. (18). From Theorem 2(ii) and Lemma 1, we infer

$$\begin{aligned}\lim_{k\to\infty} (\|\{f_i^{k+1}\}_{i=1}^N\|_F &\le \lim_{k\to\infty} \rho_b \|z^{k+1} - c^k\|_F \\ &= 0. \end{aligned} \quad (32)$$

Let

$$z_i^k = A_i^k. \quad (33)$$

From Theorem 2(i), we know that

$$\begin{aligned}\lim_{k\to\infty} J(\left\{z_i^{k+1}\right\}_{i=1}^{j-1}, z_j^{k+1}, \left\{z_i^k\right\}_{i=j+1}^N) &= \lim_{k\to\infty} J(z^k) \\ &= J(z^*). \end{aligned} \quad (34)$$

Therefore, from Eq. (31), Eq. (32), Eq. (33) and Eq. (34), we have

$$\begin{aligned}0 &\in \partial_{z_j} \lim_{k\to\infty} J(\left\{z_i^{k+1}\right\}_{i=1}^{j-1}, z_j^{k+1}, \left\{z_i^k\right\}_{i=j+1}^N) \\ &= \partial_{z_j} J(z^*).\end{aligned}$$

This means $0 \in \partial J(z^*)$. □

E. Proof of Theorem 4

Proof. From Theorem 1, we know that $\frac{\alpha}{2}\operatorname{Tr}((A_N)^T L A_N)$ is an infinitely differentiable function, and the norm of its derivatives of any order is also continuous, so $\frac{\alpha}{2}\operatorname{Tr}((A_N)^T L A_N)$ is a real analytic function (real analytic functions are all KŁ functions [23]). Since the Frobenius norm, ℓ_p-norm and $\ell_{2,1}$-norm are also all KŁ functions [17], it follows that Eq. (7) is a KŁ function. Therefore, from Definition 6, there exists a concave function ϕ so that

$$\phi^{'}(J(z^k) - J(\overline{z}))dist(0, \partial J(z^k)) \ge 1. \quad (35)$$

From ϕ is the convex function, we have

$$\begin{aligned}\phi(J(z^{k+1}) - J(\overline{z})) &\le \phi(J(z^k) - J(\overline{z})) \\ &+ \phi^{'}(J(z^k) - J(\overline{z}))(J(z^{k+1}) - J(z^k)).\end{aligned} \quad (36)$$

From Lemma 1 and Theorem 3, we infer

$$dist(0, \partial J(z^k)) \le \rho_b \|z^k - c^{k-1}\|_F. \quad (37)$$

Since the Eq. (35) and Eq. (37), we infer

$$\begin{aligned}\phi^{'}(J(z^k) - J(\overline{z})) &\ge \frac{1}{dist(0, \partial J(z^k))} \\ &\ge \frac{1}{\rho_b \|z^k - c^{k-1}\|_F}.\end{aligned} \quad (38)$$

Let $J(k) = J(z^k) - J(\overline{z})$, from Eq. (36), Eq. (37), Eq. (38) and Theorem 2(i), we have

$$\begin{aligned}\phi(J(k)) - \phi(J(k+1)) &\ge \phi^{'}(J(k))(J(k) - J(k+1)) \\ &\ge \frac{J(k) - J(k+1)}{\rho_b \|z^k - c^{k-1}\|_F} \\ &\ge \frac{\rho \|z^{k+1} - c^k\|_F^2}{\rho_b \|z^k - c^{k-1}\|_F}.\end{aligned}$$

Define C=$\frac{\rho}{\rho_b}$, C is a constant, so we infer

$$\|z^{k+1} - c^k\|_F^2 \le C(\phi(J(k)) - \phi(J(k+1)))\|z^k - c^{k-1}\|_F.$$

Using the fact that $2ab \le a^2 + b^2$

$$2\|z^{k+1} - c^k\|_F \le C(\phi(J(k)) - \phi(J(k+1))) + \|z^k - c^{k-1}\|_F.$$

Sum both sides

$$\begin{aligned}2\sum_{k=l+1}^K \|z^{k+1} - c^k\|_F &\le \sum_{k=l+1}^K \|z^k - c^{k-1}\|_F \\ &+ C(\phi(J(l+1)) - \phi(J(K+1))) \\ &= C(\phi(J(l+1)) - \phi(J(K+1))) \\ &+ \|z^{l+1} - c^l\|_F + \sum_{k=l+1}^K \|z^{k+1} - c^k\|_F.\end{aligned} \quad (39)$$

From Eq. (39), we can get that

$$\begin{aligned}\lim_{K\to\infty} \sum_{k=l+1}^K \|z^{k+1} - c^k\|_F &\le \|z^{l+1} - c^l\|_F + C\phi(J(l+1)) \\ &- \lim_{K\to\infty} C\phi(J(K+1)) \\ &< \infty.\end{aligned}$$

Thus we have

$$\sum_{k=l+1}^\infty \|z^{k+1} - c^k\|_F < \infty. \quad (40)$$

From Algorithm 1, no matter $c^k = z^k + \beta_k(z^k - z^{k-1})$ or $c^k = z^k$, we always have

$$\|z^{k+1} - z^k\|_F - \beta_k\|z^k - z^{k-1}\|_F \ \leq \|z^{k+1} - c^k\|_F. \quad (41)$$

From Eq. (40) and Eq. (41), we know that

$$\sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_k\|z^k - z^{k-1}\|_F) \leq \sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F < \infty.$$

Since

$$\|z^{k+1} - z^k\|_F - \beta_{max}\|z^k - z^{k-1}\|_F \leq \|z^{k+1} - z^k\|_F - \beta_k\|z^k - z^{k-1}\|_F,$$

we also have

$$\begin{aligned} &\sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_{max}\|z^k - z^{k-1}\|_F) \\ &\leq \sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_k\|z^k - z^{k-1}\|_F) \\ &< \infty, \end{aligned} \quad (42)$$

and

$$\begin{aligned} &\sum_{k=l+1}^{\infty} \|z^{k+1} - z^k\|_F - \sum_{k=l+1}^{\infty} \beta_{max}\|z^k - z^{k-1}\|_F \\ &= \sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_{max}\|z^{k+1} - z^k\|_F) \\ &+ \beta_{max}\|z^{l+1} - z^l\|_F \\ &= \sum_{k=l+1}^{\infty} (1 - \beta_{max})\|z^{k+1} - z^k\|_F \\ &+ \beta_{max}\|z^{l+1} - z^l\|_F. \end{aligned} \quad (43)$$

Let $s^k = \|z^{k+1} - z^k\|_F$, from Eq. (42) and Eq. (43), we have

$$\begin{aligned} \sum_{k=l+1}^{\infty} (1 - \beta_{max})s^k - \beta_{max}s^l &= \sum_{k=l}^{\infty}(s^{k+1} - \beta_{max}s^k) \\ &\leq \sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F. \\ &< \infty. \end{aligned}$$

Thus we infer

$$\sum_{k=l+1}^{\infty} (1 - \beta_{max})s^k < \infty. \quad (44)$$

From $(1 - \beta_{max})$ is a positive constant and Eq. (44), we infer

$$\sum_{k=l+1}^{\infty} \|z^{k+1} - z^k\|_F = \sum_{k=l+1}^{\infty} s^k < \infty.$$

Thus we have

$$\sum_{k=0}^{\infty} \|z^{k+1} - z^k\|_F = \sum_{k=0}^{\infty} s^k < \infty.$$

This shows that

$$\begin{aligned} \lim_{K\to\infty} \|z^{K+p} - z^K\|_F &= \lim_{K\to\infty} \|\sum_{k=K+1}^{K+p} (z^{k+1} - z^k)\|_F \\ &\leq \lim_{K\to\infty} \sum_{k=K+1}^{K+p} \|z^{k+1} - z^k\|_F \\ &= \lim_{K\to\infty} \sum_{k=K+1}^{\infty} s^k \\ &= 0. \end{aligned} \quad (45)$$

This means the Theorem 4 is true. □

Remark 2. *If we let F_i in Eq. (7) be defined as*

$$F_i = \lambda_i\|A_i\|_1 + \mathbb{I}_{A_i \geq 0}(A_i), \quad (46)$$

where $\lambda_i \in \mathbb{R}_+$ ($\forall i \in [N]$). Then we can use the proposed Algorithm 1 to solve ℓ_1-SGDNMF (Eq. (6)), and all theorems in Section V also hold for ℓ_1-SGDNMF. This is because the proof process in Section V only requires that F_i be lower semi-continuous KŁ functions (Definition 3 and Definition 6) with a lower bound, F_i in Eq. (46) satisfy these requirements.

F. Proof of Theorem 5

Proof. Let $B_i = \prod_{j=1}^{i-1} A_j$, $C_i = \prod_{j=i+1}^{N} A_j$, the computational complexity of B_i reachs $\mathcal{O}(m_1 \sum_{j=1}^{i-2} m_j m_{j+1})$, and the computational complexity of C_i reachs $\mathcal{O}(R \sum_{j=i+1}^{N-1} m_j m_{j+1})$. Therefore, the computational complexity of $(B_i)^T B_i A_i C_i (C_i)^T$ is

$$\begin{aligned} &\mathcal{O}(R \sum_{j=i+1}^{N-1} m_j m_{j+1} + m_1 \sum_{j=1}^{i-2} m_j m_{j+1} \\ &+ m_i^2 m_{i+1} + m_i m_{i+1}^2) \end{aligned} \quad (47)$$

Considering that B_i and C_i have been pre-calculated, the time complexity of $(B_i)^T X (C_i)^T$

$$\mathcal{O}(m_1 m_i R + m_i m_{i+1} R). \quad (48)$$

Therefore, from Eq. (47) and Eq. (48), we know that the computational cost of $\nabla_{A_i} H(A_{(N)})$ is

$$\begin{aligned} &\mathcal{O}(R \sum_{j=i+1}^{N-1} m_j m_{j+1} + m_1 \sum_{j=1}^{i-2} m_j m_{j+1} \\ &+ m_i^2 m_{i+1} + m_i m_{i+1}^2 + m_1 m_i R + m_i m_{i+1} R). \end{aligned} \quad (49)$$

The cost of computing the Lipschitz constant, projection to nonnegative, and objective evaluation is negligible compared to the cost of computing the partial gradient $\nabla_{A_i} H(A_{(N)})$. Therefore, the time complexity of the Algorithm 1 in each iteration is approximately estimated as

$$\begin{aligned} &\mathcal{O}(\sum_{i=1}^{N-2}(R \sum_{j=i+1}^{N-1} m_j m_{j+1} + m_1) + \sum_{i=3}^{N}(\sum_{j=1}^{i-2} m_j m_{j+1}) \\ &+ \sum_{i=1}^{N}(m_i^2 m_{i+1} + m_i m_{i+1}^2 + m_1 m_i R + m_i m_{i+1} R)). \end{aligned} \quad (50)$$

□